

Sketch-to-Image Translation Using Pix2Pix

Flower Image Generation from Sketches

Advanced Topics in Deep Learning

Final Project

Students

Angel Sasin

Asher Pe'er

Institution

The College of Management Academic Studies

1 Abstract

This project focuses on sketch-to-image translation in the flower domain. Given a 128×128 sketch image as input, the goal is to generate a realistic 128×128 RGB flower image. We implemented and compared two approaches: a naive encoder-decoder baseline and a Pix2Pix conditional GAN. The Pix2Pix model used a U-Net generator and a PatchGAN discriminator. Experiments included different PatchGAN depths and different values of the L1 loss weight λ . The final model was selected mainly according to visual quality together with quantitative evaluation.

2 Introduction

Image-to-image translation is the task of converting an image from one visual representation into another. In this project, the input is a sketch of a flower, and the output is a realistic RGB flower image.

The project is based on Pix2Pix, which uses conditional Generative Adversarial Networks (cGANs) for supervised image-to-image translation. Unlike a simple reconstruction model that only minimizes pixel-level error, Pix2Pix combines reconstruction loss with adversarial learning. This helps the model generate outputs that are both close to the target image and visually more realistic.

3 Dataset and Preprocessing

The dataset used in this project is Oxford Flowers 102. Each original RGB image was resized to 128×128 pixels according to the project requirements. For each image, a sketch version was generated using the sketch-generation process provided in the assignment instructions.

This created paired training data:

- Input: sketch image, shape: $128 \times 128 \times 1$
- Target: RGB flower image, shape: $128 \times 128 \times 3$

Both the sketch input and the RGB target image were normalized to the range $[-1, 1]$. This normalization matches the final `tanh` activation of the generator and helps stabilize training.

4 Data Pipeline

The data pipeline was implemented using TensorFlow Datasets and TensorFlow data operations. The process included loading the Oxford Flowers 102 dataset, resizing images, converting them into sketch-target pairs, normalizing the values, batching the data, and using prefetching for efficient training.

The dataset was divided into training, validation, and test sets. The training set was used for model learning, the validation set was used for monitoring performance and early stopping, and the test set was used for final evaluation.

5 Model Architectures

5.1 Naive Encoder-Decoder Baseline

The naive baseline receives a sketch image and directly predicts the RGB flower image. It follows a simple encoder-decoder structure. The encoder reduces the spatial resolution and extracts features, the bottleneck stores a compressed representation, and the decoder upsamples the representation back to the original image size.

The model was trained using pixel-wise reconstruction loss. The main limitation is that pixel-wise losses often produce blurry outputs because the model averages possible color and texture solutions.

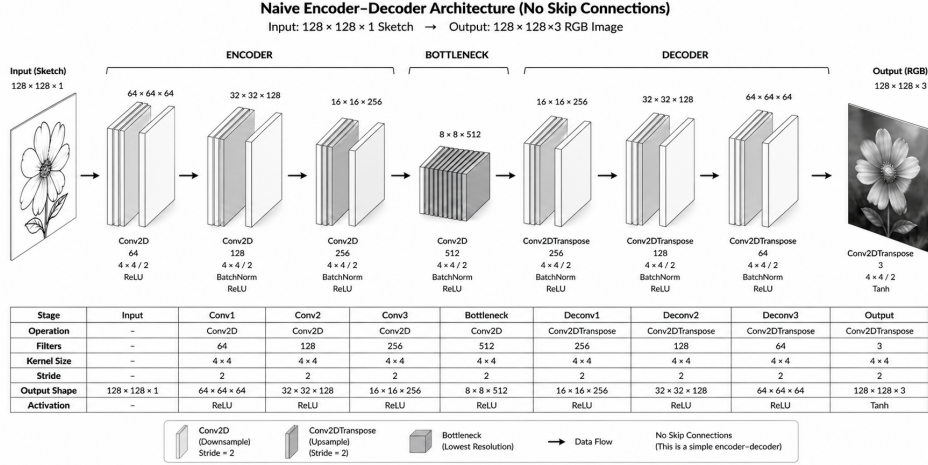


Figure 1: Naive encoder-decoder architecture without skip connections.

5.2 Pix2Pix Conditional GAN

Pix2Pix is a conditional GAN designed for supervised image-to-image translation. It consists of two networks: a generator and a discriminator.

GANs introduce an adversarial training process. The generator attempts to synthesize realistic images, while the discriminator attempts to distinguish generated images from real images. This adversarial process encourages sharper and more realistic outputs.

The generator receives the sketch image and produces an RGB flower image. In this project, the generator is based on a U-Net architecture with skip connections. These skip connections help preserve spatial information from the input sketch and improve structural consistency.

The discriminator receives image pairs and learns to distinguish between real pairs and generated pairs. Instead of classifying the whole image as real or fake, the PatchGAN discriminator classifies local patches. This encourages realistic local textures.

The generator loss combines adversarial loss and L1 reconstruction loss:

$$G^* = \arg \min_G \max_D L_{cGAN}(G, D) + \lambda L_{L1}(G) \quad (1)$$

The adversarial loss encourages realism, while the L1 loss keeps the generated image close to the target image.

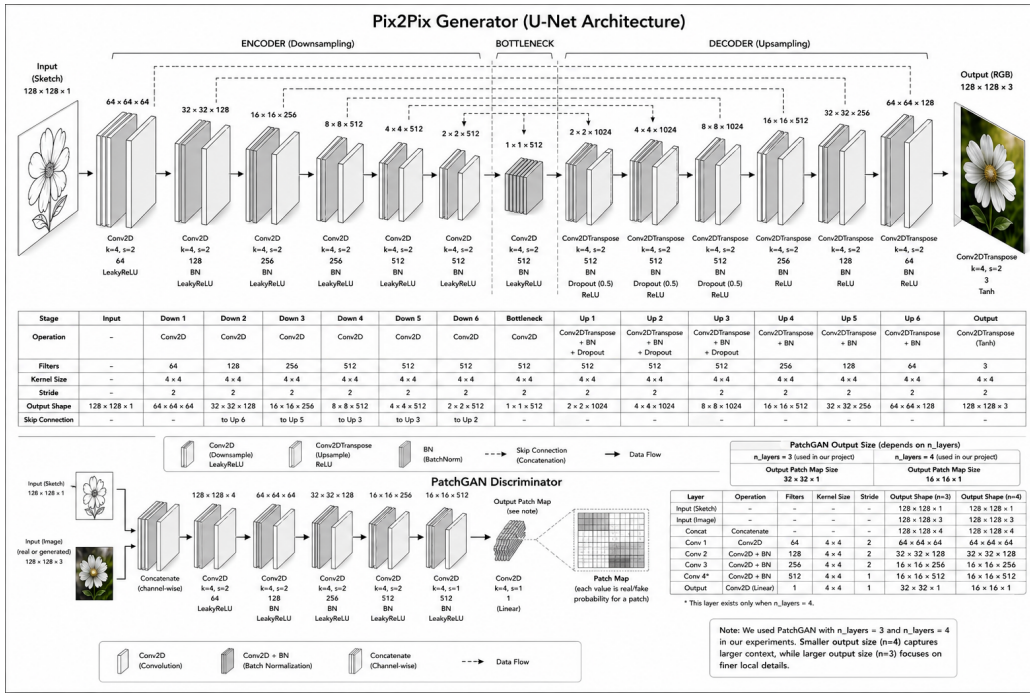


Figure 2: Pix2Pix U-Net generator and PatchGAN discriminator.

6 Experiments

6.1 Training Setup

All models were implemented using Python and TensorFlow Keras. The main hyperparameters are summarized below.

Parameter	Value
Image Size	128 × 128
Batch Size	16
Learning Rate	0.0002
Optimizer	Adam
λ Values	50, 100, 200
PatchGAN Variants	$n = 2, n = 3, n = 4$
Generator Output Activation	Tanh

Table 1: Main training hyperparameters used throughout the experiments.

6.2 Reconstruction Baseline Experiment

The first experiment evaluated the naive encoder-decoder model trained only with reconstruction loss. This model served as a baseline for understanding the limitations of reconstruction-based image generation without adversarial supervision.

The model was able to reconstruct the general structure and approximate color distribution of the flower images. However, the generated outputs often appeared blurry and lacked realistic

high-frequency texture details, which is a common limitation of pixel-wise reconstruction objectives.

6.3 PatchGAN Receptive Field Experiment

To evaluate the effect of the PatchGAN discriminator, we trained Pix2Pix models with different discriminator depths. Different depths change the receptive field size, meaning how much of the image each discriminator decision observes.

A smaller PatchGAN focuses more on local texture details, while a larger PatchGAN observes a wider region and may improve global consistency.

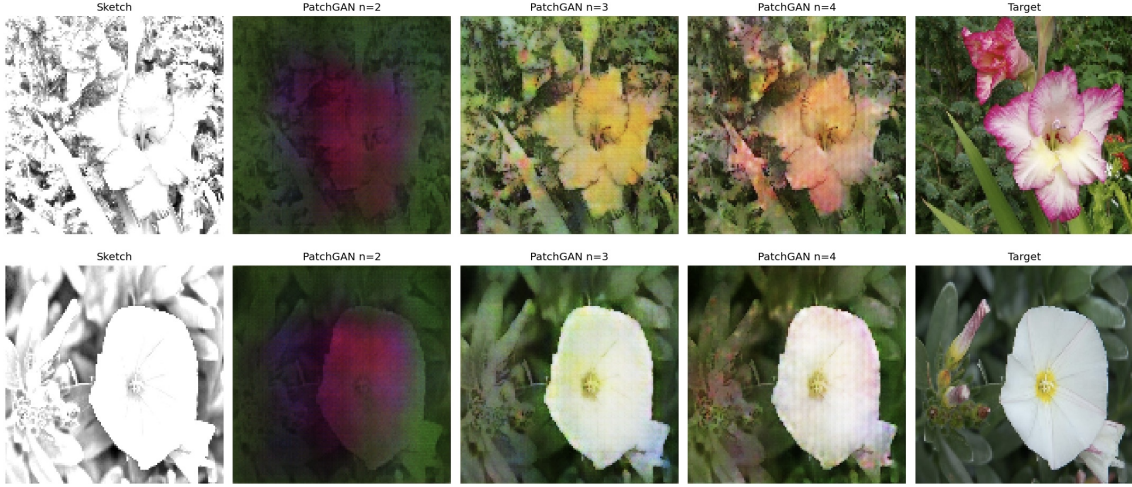


Figure 3: PatchGAN comparison using different discriminator depths. The $n = 2$ model was less stable, while $n = 3$ and $n = 4$ produced more realistic results.

6.4 Lambda Experiment

The Pix2Pix generator loss includes the parameter λ , which controls the weight of the L1 reconstruction loss. We tested $\lambda = 50$, $\lambda = 100$, and $\lambda = 200$ to analyze the balance between realism and similarity to the target image.



Figure 4: Effect of different λ values. $\lambda = 100$ produced the best balance between realistic texture and target similarity.

7 Results and Comparison

The evaluated models were compared using both quantitative metrics and qualitative visual analysis. The quantitative evaluation included L1/MAE, Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index Measure (SSIM).

Model	L1 / MAE	MSE	PSNR	SSIM
Naive MSE	0.2879	0.1546	14.45	0.4344
Pix2Pix $n = 3, \lambda = 100$	0.2699	0.1459	14.89	0.6066
Pix2Pix $n = 4, \lambda = 100$	0.2670	0.1444	14.90	0.5999

Table 2: Final model comparison on the test set.

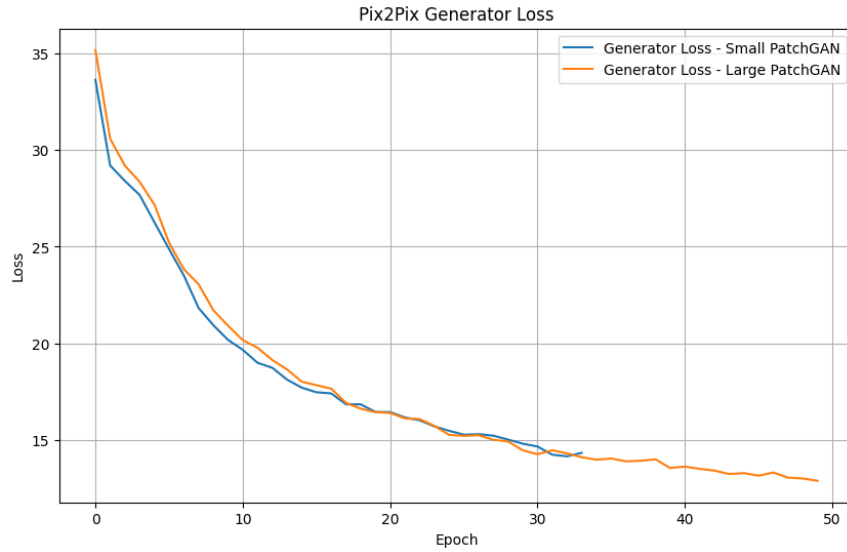


Figure 5: Pix2Pix generator loss during training for Small and Large PatchGAN configurations.



Figure 6: Final visual comparison between the naive baseline and Pix2Pix models. Pix2Pix generates sharper textures and more realistic flower structures compared to the blurry outputs of the naive baseline.

8 Discussion

The experiments performed in this project demonstrate the effectiveness of conditional adversarial training for sketch-to-image translation tasks. The naive encoder-decoder baseline was able to reconstruct coarse flower structures and approximate color distributions, but the generated outputs remained blurry and lacked realistic local textures. This limitation is mainly caused by the use of reconstruction-based losses alone, which tend to average multiple possible image solutions.

In contrast, the Pix2Pix conditional GAN significantly improved perceptual realism and structural consistency. The adversarial learning process encouraged the generator to synthesize sharper textures, more realistic flower details, and improved color consistency compared to reconstruction-only approaches.

The experiments also showed that discriminator receptive field size affects the balance between local texture quality and global image consistency. In addition, different λ values influenced the trade-off between reconstruction similarity and perceptual realism. Among the evaluated configurations, $\lambda = 100$ provided the most balanced overall results.

Despite the strong results achieved by the Pix2Pix models, some generated outputs still contained artifacts or imperfect texture regions, especially for more complex flower structures. Future work could explore higher image resolutions, perceptual losses, attention mechanisms, or more advanced generative architectures in order to further improve image quality and realism.

9 Conclusion

This project demonstrated that Pix2Pix improves sketch-to-image translation compared to a naive encoder-decoder baseline.

Although the $n = 4$ model achieved slightly better quantitative metrics, the visual outputs of the $n = 3$ model were sharper and more visually consistent across different flowers. Therefore, the $n = 3$ configuration was selected as the final practical model. Overall, the project demonstrates the effectiveness of conditional GAN architectures for realistic sketch-to-image translation tasks.

10 How to Run the Notebooks

Training Notebook: Open the training notebook and run all cells. The notebook loads the dataset, creates sketch-target pairs, trains the naive and Pix2Pix models, saves the trained generators, and displays training outputs and loss graphs.

Test Notebook: Open the test notebook and run all cells. The notebook automatically loads the trained Pix2Pix generator and allows the user to upload a sketch image. The generated RGB flower output is then displayed using the trained model.

11 References

- Isola, P., Zhu, J.-Y., Zhou, T., and Efros, A. A. Image-to-Image Translation with Conditional Adversarial Networks. CVPR, 2017.
- Oxford Flowers 102 Dataset. TensorFlow Datasets Catalog.